

Drug repurposing in kidney disease



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Drug repurposing, is the re-tasking of known medications for new clinical indications. Advantages, compared to *de novo* drug development, include reduced cost and time to market plus the added benefit of a known pharmacokinetic and safety profiles. Suitable drug candidates are identified through serendipitous observations, data mining, or increased understanding of disease mechanisms. This review highlights drugs suited for repurposing in kidney disease. The main cause of mortality in patients with chronic kidney disease is cardiovascular disease. Hence, we have included CV endpoints for the drugs. This review begins with candidates in acute kidney injury: vasodilators levosimendan and vitamin D, followed by candidates in CKD, with particular focus on diabetic kidney disease, autosomal dominant polycystic kidney disease, and focal segmental glomerulosclerosis. Examples include glucose-lowering drugs (sodium glucose co-transporter 2 inhibitors, glucagon-like peptide 1 agonists, and metformin), which have mechanistic potential for cardiac and/or renal protection beyond glucose lowering, with broader applicability to the nondiabetic population; xanthine oxidase inhibitors (allopurinol, febuxostat), selective endothelin receptor A antagonist (atrasentan), Janus kinase inhibitor (baricitinib), selective costimulation modulator (abatacept), pentoxifylline, and the DNA demethylating agent/vasodilator (hydralazine).

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Drug repurposing is a cost-effective pragmatic strategy to treat patients with kidney disease. Often the pharmacokinetic, pharmacodynamics, and safety profile is already known. This review focuses on potential suitable candidates for drug repurposing in acute kidney injury (AKI) and chronic kidney disease (CKD).

The International Diabetes Federation estimated in 2007 that the number of people with diabetes will reach 380 million in 2025. The startling fact is that in 2015 this figure was already 415 million worldwide. In high-income countries, diabetes is the leading cause of cardiovascular (CV) disease and kidney failure, consuming 12% of global health expenditure.¹ There are clear barriers which have limited drug development in this area, despite a wealth of relevant research, leaving translatable therapeutics an area of significant unmet needs. These include cost, time to hard endpoints, and targets that are not easily reached by drugs. Ideally *de novo* drug development should occur in parallel with drug repurposing.

Screening for novel targets often involves large data mining where genes or proteins are screened for differences between diseased and nondiseased states.² Identified targets require validation to determine if the differences are associative or causative. Once proof of concept is established preclinically, it is followed by a lengthy process of preclinical toxicity testing and establishing pharmacokinetic and dynamic profiles in humans before it can enter into phase 2 trials, which provide interim endpoints to support a costly phase 3 trial. If endpoints are positive, this allows entry into the commercially competitive market. The Tufts Centre for the Study of Drug Development recently published a study that estimated the cost to develop and gain marketing approval for a new drug to be approximately \$2.6 billion.³ Clearly, costs to the pharmaceutical industry are compounded by the large number of targets that “fall over” or do not make it at multiple points in the development process.

In drug repurposing, a similar approach to data mining coupled with bioinformatics and preclinical studies can identify whether an existing drug can be used for other indications. The impetus to repurpose can be based on serendipitous observations during preclinical studies or clinical use. As for novel drugs, new indications must be rigorously tested in clinical trials. The main advantages of repurposing are the known side effects and/or toxicities, which inevitably mean less cost, less time, and a better safety profile (Figure 1).

One of the best examples of drug repurposing is thalidomide. Used as an over-the-counter antiemetic in the 1950s, it was subsequently withdrawn after it was recognized as the cause of horrific birth defects. Thalidomide was reintroduced

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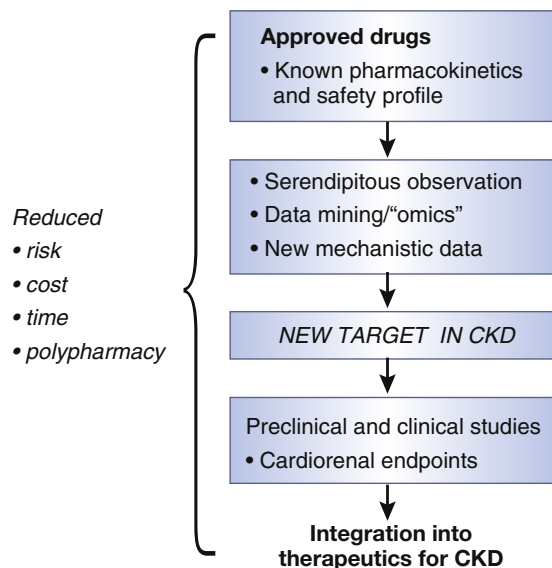


Figure 1 | Summary of drug repurposing pathway. This diagram summarizes the pathway of drug repurposing in chronic kidney disease. It highlights the advantages over *de novo* drug development.

under strict guidelines for new indications and approved by the US Food and Drug Administration in 1998 for use in treating leprosy and again in 2006 for treating multiple myeloma.

In this review, we highlight examples of drugs that, when repurposed, may be useful to treat both AKI and CKD. As the main cause of mortality in CKD is CV in its origin, it is important that both CV and renal endpoints are addressed. Most recent trials are designed for safety and use as the primary outcome “major adverse cardiac events.” For this reason, the examples provided in this review are heavily focused on meeting those endpoints. In addition, drug dosages are an important consideration, given that the kidney is one of the main organs in drug elimination. Additionally, the repurposed drug is attractive if it is inexpensive and globally accessible (Figure 2).

Acute kidney injury

In contrast to previously held beliefs, it is now well recognized that AKI is independently associated with risk of death, CV events, and incident/progressive CKD,^{4,5} and is viewed as part of a continuum to CKD. As such, it is a global health priority to reduce or prevent AKI.⁶

Vasodilator: levosimendan. Levosimendan is a calcium sensitizer and adenosine triphosphate-dependent potassium (K_{ATP}) channel agonist with vasodilatory and inotropic effects originally developed for treating acute cardiac failure. Given its mechanism of action, it is not surprising that levosimendan has pleiotropic effects beyond improved cardiac function. An expert panel position paper published in 2016 highlighted potential novel applications which included kidney injury.⁷

Preclinical studies have shown that levosimendan offers renoprotection in ischemia reperfusion injury as a result of activation of the mitochondrial ATP-sensitive potassium

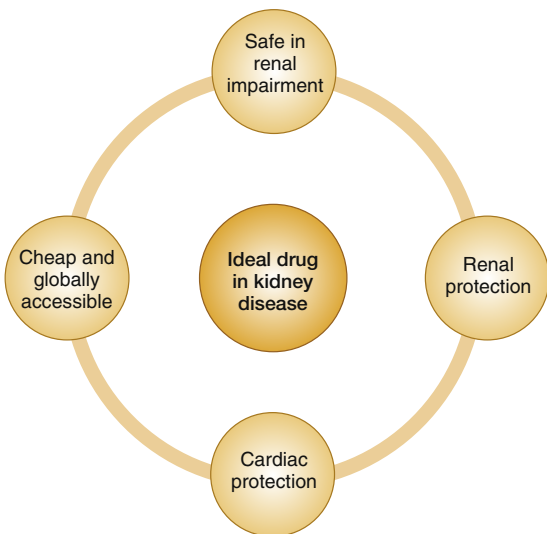


Figure 2 | Features of an ideal drug for a patient with kidney disease are highlighted.

channel and nitric oxide (NO) synthase⁸ and in experimental endotoxemic acute renal failure in mice, most likely as a result of vasodilator properties.⁹

Meta-analyses suggest that perioperative levosimendan may reduce renal injury in adult patients undergoing cardiac surgery,^{10,11} presumably due to vasodilation increasing blood flow, glomerular filtration rate (GFR), and renal oxygenation. Previous negative trials with vasodilators in this setting temper the enthusiasm with levosimendan. We cautiously await the outcome of clinical trials evaluating the efficacy of levosimendan in AKI in adult intensive care patients (ClinicalTrials.gov identifier: NCT01720030) and in patients undergoing cardiac surgery (Effects of Levosimendan in Acute Kidney Injury After Cardiac Surgery [LEVOAKI] study; NCT02531724).

Vitamin D. Vitamin D is involved in calcium homeostasis and bone metabolism. Vitamin D status also influences other systems including the immune system and CV and endothelial systems, all of which can impact kidney function.¹²

Vitamin D may reduce inflammatory and profibrotic pathways relevant to kidney disease through multiple mechanisms. Preclinical studies in rats have shown that vitamin D deficiency contributes to vascular damage, aggravating ischemic AKI.^{13,14} Vitamin D suppresses transforming growth factor-beta-mediated tubular epithelial-to-mesenchymal transition and renal fibrosis in a vitamin D receptor-dependent manner.¹⁵ Activated vitamin D receptor is also involved in translocation and activation of nuclear factor- κ B (NF- κ B) in kidney cells.¹⁶ Vitamin D pretreatment regulates renal inflammatory responses in lipopolysaccharide-induced AKI by repressing NF- κ B in the renal tubules. Lipopolysaccharide, which activates renal NF- κ B, reciprocally suppresses renal vitamin D receptor and its target gene.¹⁷ Patients with AKI manifested a marked decrease in vitamin D level at the time of AKI diagnosis, which increased with the severity of AKI. No association between serum vitamin D level at the

time of AKI diagnosis and 90-day all-cause mortality was found in clinical studies.¹⁸

The VID-AKI study (Vitamin D in Critically Ill Patients With Acute Kidney Injury; NCT02869919) aims to investigate whether there is a difference in vitamin D levels between critically ill adult patients with AKI and those without AKI. In addition, the phase 2 study Activated Vitamin D for the Prevention and Treatment of Acute Kidney Injury (ACTIVATE-AKI) study in a population of critically ill patients (NCT02962102) will assess the composite of daily serum creatinine concentration, the need for renal replacement therapy, or death within 7 days as the primary renal outcome. However, given the number of negative trials to date, the likelihood of vitamin D altering multifactorial AKI is low.

Chronic kidney disease

Addressing the need for better therapies to reduce the burden of CKD must be done in the context of concurrently addressing CV outcomes. Large-scale trials are unlikely to be conducted where patent protection has expired, often leaving this opportunity to government-supported trial programs. The focus of subsequent sections is diabetic nephropathy (DN), the leading cause of CKD.

Glucose-lowering drugs: beyond glycemic control? Recent diabetic drugs originally developed to lower glycemia have been found to also offer CV and/or renal benefit independent of glucose-lowering function. The indication for these drugs in the nondiabetic CKD population for their CV and renal benefit needs to be evaluated based on recent and upcoming clinical trial outcomes.

SGLT2 inhibitors. Sodium and glucose co-transporter 2 (SGLT2) inhibitors block the reuptake of sodium and glucose into the kidney proximal tubule (PT), leading to glycosuria and natriuresis. The SGLT2 inhibitor empagliflozin has been found in the BI 10773 (Empagliflozin) Cardiovascular Outcome Event Trial in Type 2 Diabetes Mellitus Patients (EMPA-REG OUTCOME; NCT01131676) to significantly lower the relative risk of CV and all-cause mortality by 35%, as well as the composite secondary microvascular endpoints by 38%, which were mainly renal in nature, a landmark finding in renoprotection.^{19,20} This could be explained only partly by the favorable metabolic effects, namely reduction in HbA_{1c}, in blood pressure, and in body weight. Above and beyond glucose-lowering is the direct reduction of sodium reuptake at the level of the kidney PT. This results in a contracted intravascular volume and a reduction in glomerular pressure through tubuloglomerular feedback activation. In addition, there are alterations in intrarenal hormonal activation, uric acid (UA) transport, and lipid metabolism. These findings have now led to evaluation of empagliflozin for the treatment of heart failure with diabetes (Empagliflozin in Heart Failure: Diuretic and Cardio-Renal Effects [EMPA] study; NCT03027960) and without diabetes (Empagliflozin Impact on Hemodynamics in Patients With Diabetes and Heart Failure [EMBRACE-HF]; NCT03030222). In December 2016, the US Food and Drug Administration

approved a new indication for empagliflozin (Jardiance) to reduce the risk of CV death in high-risk adult patients with type 2 diabetes (T2D), followed closely by the Australian Therapeutic Goods administration in January 2017.

Similarly, the CANagliflozin Treatment And Trial Analysis-Sulfonylurea (CANTATA-SU) study showed that the decline in estimated GFR (eGFR) was significantly slower with canagliflozin. Results were adjusted for differences in HbA_{1c}, again suggesting that the findings were independent of glucose differences. There are limitations to this study as it was a *post hoc* analysis and not powered to investigate renal effects.²¹ The CANagliflozin cardioVascular Assessment Study (CANVAS) and CANVAS-R studies (CANVAS program) showed that patients treated with canagliflozin had reductions of heart failure and renal endpoints of albuminuria and eGFR similar to those in the EMPA-REG study. However, the relative risk of CV death was significantly reduced in the EMPA-REG study but not in the CANVAS program. This may be due to the fact that the EMPA-REG cohort had a higher baseline CV risk. There was a 2-fold greater risk of amputation, primarily of the toe or metatarsal.²² Despite the differences in CV death between the studies, both of the studies demonstrated similar reduction in renal endpoints. However, there are also other important differences which are not yet fully understood. Ongoing trials will provide further comparison. The CREDENCE study (Evaluation of the Effects of Canagliflozin on Renal and Cardiovascular Outcomes in Participants With Diabetic Nephropathy; NCT02065791) will evaluate the primary composite endpoint of end-stage kidney disease, doubling of serum creatinine, renal or CV death in 4461 patients with DN. The DECLARE-TIMI58 (Multicenter Trial to Evaluate the Effect of Dapagliflozin on the Incidence of Cardiovascular Events; NCT01730534) study has a large population of primary care patients with the endpoints of the incidence of CV death, myocardial infarction, or ischemic stroke in patients with T2D. Results are expected in 2019.

Cardiorenal protection independent of glucose-lowering is relevant to nondiabetic CKD, and this is reflected in the number of clinical trials in this population. The EMPRA (EMPagliflozin and RAs in Kidney Disease; NCT03078101) study is a prospective, pilot study in CKD patients to show whether empagliflozin in addition to angiotensin-converting enzyme inhibitor treatment significantly increases Ang-1-7 levels compared to angiotensin-converting enzyme inhibitor treatment alone, with renal parameters as a secondary endpoint. A Study to Evaluate the Effect of Dapagliflozin on Renal Outcomes and CV Mortality in Patients With CKD (Dapa-CKD) has just commenced (NCT03036150). In addition, the Effects of Dapagliflozin in non-diabetic Patients With Proteinuria (DIAMOND; NCT03190694) study aims to evaluate the change in 24-hour proteinuria with dapagliflozin for 6 weeks relative to placebo. Treating to Reduce Albuminuria and Normalize Hemodynamic Function in Focal ScLerosis With dApagliflozin Trial Effects (TRANSLATE; NCT02585804) is another recently commenced trial to determine whether the addition of SGLT2 inhibitors to

standard of care leads to reduced intraglomerular pressure and suppression of proteinuria in patients with focal segmental glomerular sclerosis (FSGS).

In summary, the magnitude of cardiorenal benefit seen with SGLT2 inhibitors seems greater than expected from glycemic control alone. Furthermore, there may be molecule-specific effects within this class of drug that may explain adverse effects such as amputations. Whether cardiorenal protection is also seen in the nondiabetic CKD population is currently being evaluated in clinical trials.

Glucagon like peptide: 1 receptor agonists. Glucagon-like peptide-1 (GLP-1) acts on the GLP-1 receptor in the pancreas in a glucose-dependent manner to stimulate insulin release and suppress glucagon secretion with the net effect of regulating postprandial glucose excursions. The very short half-life is a result of rapid degradation by dipeptidyl-peptidase 4 (DPP4) which cleaves 2 amino acids at the amino terminal end of peptides that have a proline (or less commonly an alanine) at the penultimate position. Hence GLP-1 (7–36) amide is cleaved to GLP-1 (9–36) amide, which is the major circulating form. Although GLP-1 (9–36) does not have insulinotropic effects, this and a shorter truncated amide GLP-1 (28–36) can contribute to pleiotropic effects, especially on the CV system.^{23,24} This is an important therapeutic consideration and is summarized in Figure 3.

GLP-1R agonist treatment is renoprotective in animal models for both T1D and T2D, most likely through its action on the glomerular endothelial and infiltrating inflammatory cells.^{25,26} The anti-inflammatory effect of GLP-1 can also be directly on mesangial cells by the GLP-1R and by reducing the receptor for advanced glycation end products (RAGE) expression through the activation of the cyclic adenosine monophosphate pathway.²⁷ The antihypertensive effects of exendin-4 and recombinant GLP-1 are thought to be through attenuation of the actions of angiotensin II,²⁸ resulting in a reduction of proteinuria and renal and CV pathology.²⁹ Similar effects were seen with the exanatide analog AC3174 in Dahl salt-sensitive rats.³⁰ Other renal effects include modulation of sodium homeostasis through its action on PT Na⁺-H⁺ exchanger isoform 3 (NHE3).^{31,32} This has been shown with i.v. infusions of GLP-1, which enhanced sodium excretion, reduced H⁺ secretion, and reduced glomerular

hyperfiltration in obese men, supporting its action at the level of the kidney PT.³³

Preclinical and clinical studies also support a role for GLP-1R agonist in CV protection. Treatment with GLP-1 pre-ischemia was able to significantly reduce infarct size in both an isolated perfused rat heart and whole-animal models of ischemia-reperfusion.³⁴ Patients with ischemia³⁵ and left ventricular dysfunction³⁶ or left ventricular dysfunction alone³⁷ who received infusion of GLP-1 showed improvement in myocardial salvage or left ventricular wall motion. The effects of GLP-1 or its analogs on CV outcomes relate to a direct effect on cardiomyocytes mediated both by GLP-1 increased myocardial insulin sensitivity and subsequent glucose uptake³⁸ and by GLP-1 binding with GLP-1R to modulate apoptotic control of the survival of these cells.^{34,39} In addition to cardiomyocytes, GLP-1R expression has been demonstrated in endothelial cells and vascular smooth muscle cells (VSMC).⁴⁰ Animal studies have shown that GLP-1 can induce NO-mediated endothelial-dependent relaxation of pulmonary artery vessel rings,^{41,42} whereas other studies have observed vasodilation independent of NO⁴³ by a direct action on VSMC through GLP-1R. These experiments have been expanded further to explore the molecular basis behind these findings, with coronary flow improvements appearing to occur through the GLP-1 metabolite GLP-1 (9–36) and at least in part through a GLP-1R-independent effect.⁴⁴ In terms of vascular cell growth, *in vivo* studies using a vascular injury model have shown that continuous infusion of exendin-4, a GLP-1 analog, reduced neointimal formation independent of metabolic effects, by suppressing platelet derived growth factor-induced proliferation of VSMC.⁴⁵ In addition, exendin-4 stimulated proliferation of human coronary artery endothelial cells by endothelial NO synthase, protein kinase A, and phosphatidylinositol-3-kinase/Akt-dependent pathways⁴⁶ and the proliferation of the vasculoprotective endothelial progenitor cells via an action on vascular endothelial growth factor.⁴⁷ The production of proinflammatory markers by endothelial cells has been altered by treatment with GLP-1 analogs. Exanatide has been associated with a significant attenuation of tumor necrosis factor- α (TNF- α) mRNA and monocyte chemoattractant protein-1 (MCP-1),⁴⁰ all of which have been associated with the development of atherosclerosis. GLP-1 and GLP-1 analogs have therefore been demonstrated to alter vascular cell growth and function and are linked to alterations in the inflammatory milieu of a number of cell types.

There are 4 completed clinical trials to date that have examined GLP-1 RA and CV outcomes. The Lixisenatide in Patients with T2D and Acute Coronary Syndrome (ELIXA) study with a primary composite endpoint of CV death, myocardial infarction, stroke, or hospitalization for unstable angina showed the noninferiority of lixisenatide to placebo ($P < 0.001$) but did not show superiority ($P = 0.81$).⁴⁸ Similarly, the Exenatide Study of CV Event Lowering Trial (EXSCLE) study showed that cardiovascular events did not differ significantly between patients who received exenatide and those who received placebo.⁴⁹

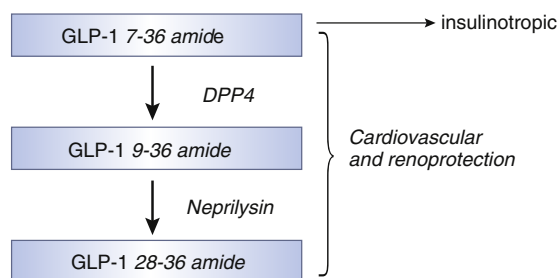


Figure 3 | Biological actions of different glucagon-like peptide (GLP) amides. Dipeptidyl peptidase 4 (DPP4) cleaves GLP 7–36 amide to GLP 9–36 amide, the major circulating form, which is then cleaved further by neprilysin.

In contrast, both liraglutide and semaglutide have been shown to offer CV and renal protection. In the Liraglutide Effect and Action in Diabetes: Evaluation of Cardiovascular Outcome Results (LEADER) study, patients with T2D and high CV risk were assigned to receive either the GLP-1 analog liraglutide or placebo. In the time-to-event analysis, the rate of the first occurrence of death from CV causes, nonfatal myocardial infarction, or nonfatal stroke among patients with T2D was lower with liraglutide than with placebo. Furthermore, there was reduction of new-onset persistent proteinuria.⁵⁰ Prespecified secondary analysis showed that, when added to usual care, liraglutide resulted in lower rates of the development and progression of DN.⁵¹ A similar result was seen with semaglutide in the Trial to Evaluate Cardiovascular and Other Long-term Outcomes With Semaglutide in Subjects With Type 2 Diabetes (SUSTAIN-6) study which was primarily powered as a noninferiority study.⁵²

Metformin. Metformin, a biguanide, has been used as a first-line diabetic agent for decades. Independently of its effect on diabetes, in observational studies metformin also reduces all-cause mortality and diseases of aging.⁵³ It is inexpensive, safe, and effective and unlikely to cause weight gain or hypoglycemia. Metformin inhibits hepatic gluconeogenesis and opposes the effects of glucagon. The detailed mechanism of action is complex and still debated but thought to be primarily at the level of the mitochondrion. The signaling pathways are related to energy metabolism, specifically adenosine monophosphate activating protein kinase (AMPK) and mammalian target of rapamycin (mTOR) inhibition.

Aging is associated with renal fibrosis and eGFR decline. In human kidney PT, metformin treatment has been shown to alleviate renal cell senescence through AMPK and mTOR signaling activation.⁵⁴ In addition, AMPK, mTOR, and sirtuins play a key role in the pathogenesis of adult polycystic kidney disease (APKD). Activating AMPK using metformin slowed renal cystogenesis in animal models of APKD.⁵⁵ As such, studies are now underway evaluating metformin as a novel therapy for APKD (Metformin as a Novel Therapy for Autosomal Dominant Polycystic Kidney Disease [TAME]; NCT02656017).

Despite longstanding concerns over the risk of lactic acidosis in patients with renal impairment, the US Food and Drug Administration has recently relaxed its recommendations for using dose-adjusted metformin in patients with CKD and eGFR between 30 and 45 ml/min.⁵⁶ A recent systematic review supports this recommendation.⁵⁷ In The UK Prospective Diabetes Study (UKPDS), use of metformin in overweight patients was associated with a significant reduction of 39% in the incidence of myocardial infarction, and this effect was not associated with the dose of metformin, more effective lowering of HbA_{1c}, or effects on plasma lipids.⁵⁸

In summary, metformin offers CV benefits, and preclinical studies show mechanistic effects on signaling pathways relevant to modulating underlying CKD.⁵⁹ Interestingly, several studies have also suggested a role in reducing cancers through insulin reduction as well as through inducing energetic

stress.⁶⁰ Metformin is ideally placed to undergo clinical trials not only in APKD but also in other types of CKD.

XO inhibitors. In addition to gout, renal stones, hypertension, and CV disease, there is now emerging evidence to suggest a causal link between hyperuricemia and the progression of CKD.^{61–65} Allopurinol is a xanthine oxidase (XO) inhibitor which is inexpensive, effective in lowering serum UA, and has a well-described toxicity profile. A newer, non-purine XO inhibitor, febuxostat, is also clinically available to lower UA. UA, an end-product of purine metabolism, can activate the renin-angiotensin system (RAS) and increase the production of reactive oxygen species.^{66,67} Interestingly, SGLT2 inhibitors reduce serum UA levels,^{19,68,69} thought to be mediated by SLC2A9b, which secretes urate back into the urine in exchange for luminal glucose.⁷⁰ This uricosuric effect may be one of the several mechanisms by which SGLT2 inhibitors deliver renoprotection.

Allopurinol. Although systematic reviews and meta-analyses broadly support a renoprotective effect in lowering UA, the main conclusion was the requirement of adequately powered randomized controlled trials to answer this question.^{71,72} In a 3-year randomized parallel-controlled study in patients with T2D, allopurinol treatment improved renal function.⁷³ In addition, a prospective, randomized trial of 113 patients with eGFR <60 ml/min showed that allopurinol decreased C-reactive protein and slowed the progression of renal disease in patients with CKD at 2 years. A *post hoc* analysis of this cohort 5 years later showed that long-term allopurinol treatment may slow kidney disease progression and reduce CV risk. This study was limited by a small sample size and a single center, and was not double blind.⁷⁴ The PERL study (A Multicenter Clinical Trial of Allopurinol to Prevent Kidney Function Loss in Type 1 Diabetes; NCT02017171) is an ongoing multicenter clinical trial of allopurinol to prevent kidney function loss in T1D. In addition, CKD-FIX (A randomized Controlled trial of slowing of Kidney Disease progression From the Inhibition of XO) also using allopurinol is in progress (trial: AKTN 10.02).

Febuxostat. Preclinical studies in animals with T1D and T2D have shown similar renoprotective effects with newer nonpurine XO inhibitors such as febuxostat.^{75–77} A placebo controlled randomized trial in a single center evaluating 93 patients over 6 months showed that febuxostat slowed the decline in eGFR in CKD stages 3 and 4 compared to placebo.⁷⁸ A small randomized controlled trial of the effects of febuxostat therapy on adipokines and markers of kidney fibrosis in asymptomatic hyperuricemic patients with DN (Effects of Febuxostat on Adipokines and Kidney Disease in Diabetic Chronic Kidney Disease ; NCT01350388) showed that, although febuxostat effectively reduced plasma UA, there were no detectable effects observed for the prespecified primary (thiobarbituric acid-reactive substance in adipose tissue, adiponectin concentration, urinary transforming growth factor-beta) or secondary endpoints (plasma TNF- α , IL-6, C-reactive protein). However, hard renal endpoints were not considered in this small trial. The effect of febuxostat on

renal and systemic hemodynamic dysfunction is currently being studied in young T1D patients (The Effect of Uric Acid Lowering in Type 1 Diabetes; NCT02344602).

Selective endothelin receptor A antagonist: atrasentan. Endothelin-1 (ET-1) is a potent vasoactive peptide which contributes to hypertension, albuminuria, insulin resistance, inflammation, fibrosis, and endothelial dysfunction.⁷⁹ There are 2 receptor subtypes that mediate the actions of ET-1, known as endothelin receptor A (ETA) and endothelin receptor B (ETB). In the mid-1990s, atrasentan (AbbVie) a selective ETA antagonist was developed following the observation that men with metastatic hormone refractory prostate cancer had much higher plasma ET-1 than healthy controls or men with confined prostate cancer.⁸⁰ Contemporaneously, this selective ETA antagonist was also found to reduce blood pressure in animals models of hypertension. Although phase 3 clinical trials did not support the use of atrasentan in prostate cancer, this drug has been revisited as a potential agent to treat DN.⁸¹

Endothelin receptors promote renal tubular growth⁸² and have proinflammatory and profibrotic effects.⁸³ ETA induces greater afferent than efferent vasoconstriction, resulting in reduced GFR.⁸⁴ Inhibitors of ETA and ETB were thought to be promising additions to kidney disease therapy. The mechanistic explanation behind the clinical observation of reduced albuminuria in the trials discussed below may be related to the restoration of the glomerular endothelial glycocalyx barrier.⁸⁵

The ASCEND study showed that avosentan (ETA antagonist) reduced albuminuria when added to standard treatment in patients with T2D and overt DN. It was prematurely terminated owing to significant fluid overload and congestive heart failure, most likely because of lack of specificity (i.e., ET_A and ET_B).⁸⁶ When added to standard therapy, the effect of atrasentan, a more selective ETA antagonist on albuminuria showed that at a dose of 0.75 mg, 50% of patients had a 40% reduction of proteinuria, although eGFR did not change significantly. Peripheral edema was still present and dose-dependent.⁸⁷ Hence, atrasentan lowered albuminuria without overt signs of fluid retention at lower doses. This prompted a larger study evaluating the efficacy and safety of 2 low doses of atrasentan (0.75 and 1.25 mg/d) on albuminuria and other renal risk-related parameters in patients with DN receiving standard therapy. This confirmed that 0.75 mg of atrasentan/d markedly lowered albuminuria in T2D with DN with manageable fluid overload-related side effects. The Study of DN with Atrasentan (SONAR) is a phase 3 clinical trial involving patients with T2D and stage 2-4 CKD (NCT01858532). Using an enrichment protocol, eligible patients received atrasentan, 0.75 mg/d for 6 weeks, to determine their urinary albumin- to-creatinine ratio response and tolerance. The primary endpoint included time to doubling of serum creatinine or onset of end-stage kidney disease, as defined by need for chronic dialysis, transplantation, or death due to renal failure. Secondary endpoints included urine albumin excretion, eGFR, and CV events including CV death, myocardial infarction, and

stroke. At the time of writing, AbbVie announced the closure of the SONAR trial because the ongoing monitoring of renal events observed in the study revealed considerably fewer endpoints than expected, which will likely affect the ability to test the SONAR study hypothesis (AbbVie Pressroom Media Statement).

JAK1/JAK2 inhibitor: baricitinib. Janus kinases (JAKs) activate signal transducer and activator of transcription (STAT). It is activated in experimental models of kidney disease including diabetes.^{88,89} The use of a nonselective JAK inhibitor (AG490) significantly reduced proteinuria in diabetic rats.⁹⁰ Brosius *et al.*⁹¹ (University of Michigan) had found that dysregulation of this pathway correlated with increased DN in humans. Mechanistic studies *in vivo* confirmed the functional relevance of this pathway in DN. Serendipitously, Eli Lilly, Co. had requested Food and Drug Administration approval for the use of baricitinib in rheumatoid arthritis and through academic-industry collaboration saw the potential of repurposing this drug for DN. In a phase 2 study evaluating the efficacy of baricitinib in T2D participants with proteinuric DN already on RAS blockade, there was a reduction of albuminuria with the active inhibitor compared to placebo.⁹¹

Selective costimulation modulator: abatacept. Abatacept suppresses T-cell-mediated responses by specifically binding to co-stimulatory signaling molecules such as B7-1 (CD80) and B7-2 (CD86) on antigen-presenting cells. It is a genetically engineered protein approved in 2005 for the treatment of rheumatoid arthritis.

The preclinical observation that podocyte B7-1 is induced in a variety of disease conditions in mice was published by Reiser *et al.*⁹² in 2004. In 2013, targeting B7-1 toward clinical use of abatacept in patients with FSGS raised the possibility that this drug could be repurposed for proteinuric kidney diseases.

FSGS is a proteinuric kidney disease which often results in end-stage kidney disease. Posttransplantation recurrence is challenging to treat and causes graft loss. B7-1 staining in podocytes, highly specialized filtration barrier cells of the glomerulus, was found in 13 of 21 kidney biopsy samples from patients with proteinuric kidney disease (all those with primary FSGS were positive). B7-1 staining in podocytes was not evident in normal kidneys. Based on this and supportive *in vitro* mechanistic studies, 5 patients with treatment-resistant FSGS were treated with abatacept (4 post-transplant and 1 primary). A partial or complete remission was achieved in all 5 patients. Of note, there has been some doubt as to whether podocytes express B7, and it is unclear whether this is a technical issue.⁹³ A phase 2 pilot study is currently underway to assess the efficacy and safety of abatacept in decreasing proteinuria in patients with FSGS or minimal change disease (Pilot Study to Evaluate the Safety and Efficacy of Abatacept in Adults and Children 6 Years and Older With Excessive Loss of Protein in the Urine Due to Either Focal Segmental Glomerulosclerosis [FSGS] or Minimal Change Disease [MCD]; NCT02592798).

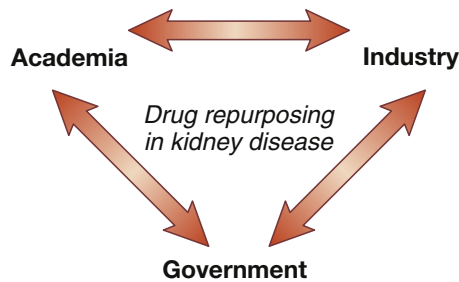


Figure 4 | Summary of the collaboration and partnership required in successful drug repurposing.

Betalcept is a second-generation version which has undergone modifications to decrease dissociation rates from B7-1 and B7-2 and is exclusively used in transplantation. This agent will not be discussed further here.

Phosphodiesterase inhibitor: pentoxifylline. Pentoxifylline is a xanthine derivative and a nonselective phosphodiesterase inhibitor with anti-inflammatory properties. Having hemorheologic effects of reducing blood viscosity and increasing microcirculatory flow, it was first approved for the treatment of intermittent claudication. Preclinical studies have shown renoprotection secondary to anti-inflammatory and antifibrotic effects.^{94–96} The PREDIAN trial (Pentoxifylline for Renoprotection in Diabetic Nephropathy) has shown that pentoxifylline when added to RAS blockade reduced the progression of nephropathy as measured by a lesser reduction of eGFR and albuminuria over 2 years.⁹⁷ This has been supported by recent metaanalyses.^{98,99}

DNA demethylating agent and vasodilator: hydralazine. Hydralazine, a vasodilator, was approved as an antihypertensive in 1953 and as a heart failure agent in the 1980s. Recently, hydralazine, at a lower dose, has been recognized as a DNA demethylating agent (i.e., able to remove an attached methyl group from nucleotides in DNA). This occurs through the induction of the endogenous demethylating enzyme Tet3.¹⁰⁰

Epigenetics is a term used to describe heritable changes in the chromatin structure without DNA sequence change. Chronic stimuli such as high glucose induce epigenetic changes either through DNA methylation or histone deacetylation, which can alter gene regulation and ultimately lead to kidney disease.

DNA methylation has been well described in population studies of CKD subjects.^{101–103} Preclinical studies support the use of low-dose hydralazine in kidney disease as a way to alter pathological gene expression. Hydralazine-induced demethylation in a murine model of AKI-to-CKD progression attenuated renal fibrosis and preserved excretory renal function independent of its blood pressure-lowering effects.¹⁰⁴

Clinical trials using hydralazine as a demethylating agent in CKD with hard cardiorenal endpoints are lacking. Epigenetic modification is an area worth exploring, given the association of histone modification with CKD.

Conclusions

Drug repurposing offers the opportunity to overcome inherent limitations in the *de novo* drug development pathway

such as prohibitive costs and time. As shown in Figure 4, a fully functional translational pipeline based on collaborative efforts between academia and industry is crucial for successful outcomes in a timely manner. This is often limited by poor economic incentive with patent expiration, leaving nonprofit organizations or governments to bridge stakeholders. In this review we have provided various classes of drugs being repurposed or suitable for repurposing in kidney disease, including those that have concomitant CV and renal benefit.

DISCLOSURE

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REFERENCES

- Ogurtsova K, da Rocha Fernandes JD, Huang Y, et al. IDF diabetes atlas: global estimates for the prevalence of diabetes for 2015 and 2040. *Diabetes Res Clin Pract.* 2017;128:40–50.
- Zhang M, Luo H, Xi Z, Rogaeva E. Drug repositioning for diabetes based on “omics” data mining. *PLoS One.* 2015;10:e0126082.
- DiMasi JA, Grabowski HG, Hansen RW. Innovation in the pharmaceutical industry: new estimates of R&D costs. *J Health Econ.* 2016;47:20–33.
- Chawla LS, Eggers PW, Star RA, Kimmel PL. Acute kidney injury and chronic kidney disease as interconnected syndromes. *N Engl J Med.* 2014;371:58–66.
- Coca SG, Singanamala S, Parikh CR. Chronic kidney disease after acute kidney injury: a systematic review and meta-analysis. *Kidney Int.* 2012;81:442–448.
- Levin A, Tonelli M, Bonventre J, et al. Global kidney health 2017 and beyond: a roadmap for closing gaps in care, research, and policy. *Lancet.* 2017;390:1888–1917.
- Farmakis D, Alvarez J, Gal TB, et al. Levosimendan beyond inotropy and acute heart failure: evidence of pleiotropic effects on the heart and other organs: an expert panel position paper. *Int J Cardiol.* 2016;222:303–312.
- Grossini E, Molinari C, Pollesello P, et al. Levosimendan protection against kidney ischemia/reperfusion injuries in anesthetized pigs. *Pharmacol Exp Ther.* 2012;342:376–388.
- Zager RA, Johnson AC, Lund S, et al. Levosimendan protects against experimental endotoxemic acute renal failure. *Am J Physiol Renal Physiol.* 2006;290:F1453–1462.
- Niu ZZ, Wu SM, Sun WY, et al. Perioperative levosimendan therapy is associated with a lower incidence of acute kidney injury after cardiac surgery: a meta-analysis. *J Cardiovasc Pharmacol.* 2014;63:107–112.
- Zhou C, Gong J, Chen D, et al. Levosimendan for prevention of acute kidney injury after cardiac surgery: a meta-analysis of randomized controlled trials. *Am J Kidney Dis.* 2016;67:408–416.
- Zoccali C, Curatola G, Panuccio V, et al. Paricalcitol and endothelial function in chronic kidney disease trial. *Hypertension (Dallas).* 2014;64:1005–1011.
- de Braganca AC, Volpini RA, Canale D, et al. Vitamin D deficiency aggravates ischemic acute kidney injury in rats. *Physiol Rep.* 2015;3(3). <https://doi.org/10.14814/phy2.12331>.
- de Braganca AC, Volpini RA, Mehrotra P, et al. Vitamin D deficiency contributes to vascular damage in sustained ischemic acute kidney injury. *Physiol Rep.* 2016;4(13). <https://doi.org/10.14814/phy2.12829>.
- Ito I, Waku T, Aoki M, et al. A nonclassical vitamin D receptor pathway suppresses renal fibrosis. *J Clin Invest.* 2013;123:4579–4594.
- Chen Y, Zhang J, Ge X, et al. Vitamin D receptor inhibits nuclear factor kappaB activation by interacting with I kappaB kinase beta protein. *J Biol Chem.* 2013;288:19450–19458.
- Xu S, Chen YH, Tan ZX, et al. Vitamin D3 pretreatment regulates renal inflammatory responses during lipopolysaccharide-induced acute kidney injury. *Sci Rep.* 2015;5:18687.
- Lai L, Qian J, Yang Y, et al. Is the serum vitamin D level at the time of hospital-acquired acute kidney injury diagnosis associated with prognosis? *PLoS One.* 2013;8:e64964.

19. Zinman B, Wanner C, Lachin JM, et al. Empagliflozin, cardiovascular outcomes, and mortality in type 2 diabetes. *N Engl J Med*. 2015;373:2117–2128.
20. Wanner C, Inzucchi SE, Lachin JM, et al. Empagliflozin and progression of kidney disease in type 2 diabetes. *N Engl J Med*. 2016;375:323–334.
21. Heerspink HJ, Desai M, Jardine M, et al. Canagliflozin slows progression of renal function decline independently of glycemic effects. *J Am Soc Nephrol*. 2017;28:368–375.
22. Neal B, Perkovic V, Mahaffey KW, et al. Canagliflozin and cardiovascular and renal events in type 2 diabetes. *N Engl J Med*. 2017;377:644–657.
23. Li J, Zheng J, Wang S, et al. Cardiovascular benefits of native GLP-1 and its metabolites: an indicator for GLP-1-therapy strategies. *Front Physiol*. 2017;8:15.
24. Guglielmi V, Sbraccia P. GLP-1 receptor independent pathways: emerging beneficial effects of GLP-1 breakdown products. *Eat Weight Disord*. 2017;22:231–240.
25. Kodera R, Shikata K, Kataoka HU, et al. Glucagon-like peptide-1 receptor agonist ameliorates renal injury through its anti-inflammatory action without lowering blood glucose level in a rat model of type 1 diabetes. *Diabetologia*. 2011;54:965–978.
26. Park CW, Kim HW, Ko SH, et al. Long-term treatment of glucagon-like peptide-1 analog exendin-4 ameliorates diabetic nephropathy through improving metabolic anomalies in db/db mice. *J Am Soc Nephrol*. 2007;18:1227–1238.
27. Ishibashi Y, Nishino Y, Matsui T, et al. Glucagon-like peptide-1 suppresses advanced glycation end product-induced monocyte chemoattractant protein-1 expression in mesangial cells by reducing advanced glycation end product receptor level. *Metabolism*. 2011;60:1271–1277.
28. Hirata K, Kume S, Araki S, et al. Exendin-4 has an anti-hypertensive effect in salt-sensitive mice model. *Biochem Biophys Res Commun*. 2009;380:44–49.
29. Yu M, Moreno C, Hoagland KM, et al. Antihypertensive effect of glucagon-like peptide 1 in Dahl salt-sensitive rats. *J Hypertens*. 2003;21:1125–1135.
30. Liu Q, Adams L, Broyde A, et al. The exenatide analog AC3174 attenuates hypertension, insulin resistance, and renal dysfunction in Dahl salt-sensitive rats. *Cardiovasc Diabetol*. 2010;9:32.
31. Carraro-Lacroix LR, Malnic G, Girardi AC. Regulation of Na⁺/H⁺ exchanger NHE3 by glucagon-like peptide 1 receptor agonist exendin-4 in renal proximal tubule cells. *Am J Physiol Renal Physiol*. 2009;297:F1647–F1655.
32. Crajoinas RO, Oricchio FT, Pessoa TD, et al. Mechanisms mediating the diuretic and natriuretic actions of the incretin hormone glucagon-like peptide-1. *Am J Physiol Renal Physiol*. 2011;301:F355–F363.
33. Gutzwiller JP, Tschopp S, Bock A, et al. Glucagon-like peptide 1 induces natriuresis in healthy subjects and in insulin-resistant obese men. *J Clin Endocrinol Metab*. 2004;89:3055–3061.
34. Bose AK, Mocanu MM, Carr RD, et al. Glucagon-like peptide 1 can directly protect the heart against ischemia/reperfusion injury. *Diabetes*. 2005;54:146–151.
35. Lonborg J, Vejlsstrup N, Kelbaek H, et al. Exenatide reduces reperfusion injury in patients with ST-segment elevation myocardial infarction. *Eur Heart J*. 2012;33:1491–1499.
36. Nikolaidis LA, Mankad S, Sokos GG, et al. Effects of glucagon-like peptide-1 in patients with acute myocardial infarction and left ventricular dysfunction after successful reperfusion. *Circulation*. 2004;109:962–965.
37. Nathanson D, Ullman B, Lofstrom U, et al. Effects of intravenous exenatide in type 2 diabetic patients with congestive heart failure: a double-blind, randomized controlled clinical trial of efficacy and safety. *Diabetologia*. 2012;55:926–935.
38. Nikolaidis LA, Elahi D, Hentosz T, et al. Recombinant glucagon-like peptide-1 increases myocardial glucose uptake and improves left ventricular performance in conscious dogs with pacing-induced dilated cardiomyopathy. *Circulation*. 2004;110:955–961.
39. Timmers L, Henriques JP, de Kleijn DP, et al. Exenatide reduces infarct size and improves cardiac function in a porcine model of ischemia and reperfusion injury. *J Am Coll Cardiol*. 2009;53:501–510.
40. Arakawa M, Mita T, Azuma K, et al. Inhibition of monocyte adhesion to endothelial cells and attenuation of atherosclerotic lesion by a glucagon-like peptide-1 receptor agonist, exendin-4. *Diabetes*. 2010;59:1030–1037.
41. Golpon HA, Puechner A, Welte T, et al. Vasorelaxant effect of glucagon-like peptide-(7-36)amide and amylin on the pulmonary circulation of the rat. *Regul Pept*. 2001;102:81–86.
42. Richter G, Feddersen O, Wagner U, et al. GLP-1 stimulates secretion of macromolecules from airways and relaxes pulmonary artery. *Am J Physiol*. 1993;265(4 Pt 1):L374–L378.
43. Nystrom T, Gonon AT, Sjolholm A, Pernow J. Glucagon-like peptide-1 relaxes rat conduit arteries via an endothelium-independent mechanism. *Regul Pept*. 2005;125:173–177.
44. Ban K, Hui S, Drucker DJ, Husain M. Cardiovascular consequences of drugs used for the treatment of diabetes: potential promise of incretin-based therapies. *J Am Soc Hypertens*. 2009;3:245–259.
45. Goto H, Nomiya T, Mita T, et al. Exendin-4, a glucagon-like peptide-1 receptor agonist, reduces intimal thickening after vascular injury. *Biochem Biophys Res Commun*. 2011;405:79–84.
46. Erdogdu O, Nathanson D, Sjolholm A, et al. Exendin-4 stimulates proliferation of human coronary artery endothelial cells through eNOS-, PKA- and PI3K/Akt-dependent pathways and requires GLP-1 receptor. *Mol Cell Endocrinol*. 2010;325:26–35.
47. Xiao-Yun X, Zhao-Hui M, Ke C, et al. Glucagon-like peptide-1 improves proliferation and differentiation of endothelial progenitor cells via upregulating VEGF generation. *Med Sci Monit*. 2011;17:BR35–BR41.
48. Pfeffer MA, Claggett B, Diaz R, et al. Lixisenatide in patients with type 2 diabetes and acute coronary syndrome. *N Engl J Med*. 2015;373:2247–2257.
49. Holman RR, Bethel MA, Mentz RJ, et al. Effects of once-weekly exenatide on cardiovascular outcomes in type 2 diabetes. *N Engl J Med*. 2017;377:1228–1239.
50. Marso SP, Daniels GH, Brown-Frandsen K, et al. Liraglutide and cardiovascular outcomes in type 2 diabetes. *N Engl J Med*. 2016;375:311–322.
51. Mann JFE, Orsted DD, Brown-Frandsen K, et al. Liraglutide and renal outcomes in type 2 diabetes. *N Engl J Med*. 2017;377:839–848.
52. Marso SP, Bain SC, Consoli A, et al. Semaglutide and cardiovascular outcomes in patients with type 2 diabetes. *N Engl J Med*. 2016;375:1834–1844.
53. Campbell JM, Bellman SM, Stephenson MD, Lisy K. Metformin reduces all-cause mortality and diseases of ageing independent of its effect on diabetes control: a systematic review and meta-analysis. *Ageing Res Rev*. 2017;40:31–44.
54. Dong D, Cai GY, Ning YC, et al. Alleviation of senescence and epithelial-mesenchymal transition in aging kidney by short-term caloric restriction and caloric restriction mimetics via modulation of AMPK/mTOR signaling. *Oncotarget*. 2017;8:16109–16121.
55. Takiar V, Nishio S, Seo-Mayer P, et al. Activating AMP-activated protein kinase (AMPK) slows renal cystogenesis. *Proc Natl Acad Sci U S A*. 2011;108:2462–2467.
56. Bloomgarden Z. Diabetes: metformin and renal insufficiency is 45, or even 30, the new 60? *Nat Rev Endocrinol*. 2015;11:693–694.
57. Crowley MJ, Diamantidis CJ, McDuffie JR, et al. Clinical outcomes of metformin use in populations with chronic kidney disease, congestive heart failure, or chronic liver disease: a systematic review. *Ann Intern Med*. 2017;166:191–200.
58. Holman RR, Paul SK, Bethel MA, et al. 10-year follow-up of intensive glucose control in type 2 diabetes. *N Engl J Med*. 2008;359:1577–1589.
59. Rajani R, Pastor-Soler NM, Hallows KR. Role of AMP-activated protein kinase in kidney tubular transport, metabolism, and disease. *Curr Opin Nephrol Hypertens*. 2017;26:375–383.
60. Heckman-Stoddard BM, DeCensi A, Sahasrabudhe VV, Ford LG. Repurposing metformin for the prevention of cancer and cancer recurrence. *Diabetologia*. 2017.
61. Ficociello LH, Rosolowsky ET, Niewczas MA, et al. High-normal serum uric acid increases risk of early progressive renal function loss in type 1 diabetes: results of a 6-year follow-up. *Diabetes Care*. 2010;33:1337–1343.
62. Hovind P, Rossing P, Tarnow L, et al. Serum uric acid as a predictor for development of diabetic nephropathy in type 1 diabetes: an inception cohort study. *Diabetes*. 2009;58:1668–1671.
63. Kang DH, Nakagawa T, Feng L, et al. A role for uric acid in the progression of renal disease. *J Am Soc Nephrol*. 2002;13:2888–2897.
64. Zoppini G, Targher G, Chonchol M, et al. Serum uric acid levels and incident chronic kidney disease in patients with type 2 diabetes and preserved kidney function. *Diabetes Care*. 2012;35:99–104.
65. Mazzali M, Kanbay M, Segal MS, et al. Uric acid and hypertension: cause or effect? *Curr Rheumatol Rep*. 2010;12:108–117.
66. Mazzali M, Kanellis J, Han L, et al. Hyperuricemia induces a primary renal arteriopathy in rats by a blood pressure-independent mechanism. *Am J Physiol Renal Physiol*. 2002;282:F991–F997.

67. Corry DB, Eslami P, Yamamoto K, et al. Uric acid stimulates vascular smooth muscle cell proliferation and oxidative stress via the vascular renin-angiotensin system. *J Hypertens*. 2008;26:269–275.
68. Lytvyn Y, Skrtic M, Yang GK, et al. Glycosuria-mediated urinary uric acid excretion in patients with uncomplicated type 1 diabetes mellitus. *Am J Physiol Renal Physiol*. 2015;308:F77–F83.
69. Davies MJ, Trujillo A, Vijapurkar U, et al. Effect of canagliflozin on serum uric acid in patients with type 2 diabetes mellitus. *Diabetes Obese Metab*. 2015;17:426–429.
70. Cheeseman C. Solute carrier family 2, member 9 and uric acid homeostasis. *Curr Opin Nephrol Hypertens*. 2009;18:428–432.
71. Bose B, Badve SV, Hiremath SS, et al. Effects of uric acid-lowering therapy on renal outcomes: a systematic review and meta-analysis. *Dial Transplant*. 2014;29:406–413.
72. Kanji T, Gandhi M, Clase CM, Yang R. Urate lowering therapy to improve renal outcomes in patients with chronic kidney disease: systematic review and meta-analysis. *BMC Nephrol*. 2015;16:58.
73. Liu P, Chen Y, Wang B, et al. Allopurinol treatment improves renal function in patients with type 2 diabetes and asymptomatic hyperuricemia: 3-year randomized parallel-controlled study. *Clin Endocrinol*. 2015;83:475–482.
74. Goicoechea M, Garcia de Vinuesa S, Verdalles U, et al. Allopurinol and progression of CKD and cardiovascular events: long-term follow-up of a randomized clinical trial. *Am J Kidney Dis*. 2015;65:543–549.
75. Komers R, Xu B, Schneider J, Oyama TT. Effects of xanthine oxidase inhibition with febuxostat on the development of nephropathy in experimental type 2 Diabetes. *Br J Pharmacol*. 2016;173:2573–2588.
76. Nakamura T, Murase T, Nampei M, et al. Effects of topiroxostat and febuxostat on urinary albumin excretion and plasma xanthine oxidoreductase activity in db/db mice. *Eur J Pharmacol*. 2016;780: 224–231.
77. Lee HJ, Jeong KH, Kim YG, et al. Febuxostat ameliorates diabetic renal injury in a streptozotocin-induced diabetic rat model. *Am J Nephrol*. 2014;40:56–63.
78. Sircar D, Chatterjee S, Waikhom R, et al. Efficacy of febuxostat for slowing the GFR decline in patients with CKD and asymptomatic hyperuricemia: a 6-month, double-blind, randomized, placebo-controlled trial. *Am J Kidney Dis*. 2015;66:945–950.
79. Kohan DE, Pollock DM. Endothelin antagonists for diabetic and non-diabetic chronic kidney disease. *Br J Clin Pharmacol*. 2013;76:573–579.
80. Nelson JB, Hedican SP, George DJ, et al. Identification of endothelin-1 in the pathophysiology of metastatic adenocarcinoma of the prostate. *Nat Med*. 1995;1:944–949.
81. Carducci MA, Saad F, Abrahamsson PA, et al. A phase 3 randomized controlled trial of the efficacy and safety of atrasentan in men with metastatic hormone-refractory prostate cancer. *Cancer*. 2007;110: 1959–1966.
82. Ong AC, Jowett TP, Firth JD, et al. An endothelin-1 mediated autocrine growth loop involved in human renal tubular regeneration. *Kidney Int*. 1995;48:390–401.
83. Sarafidis PA, Lasaridis AN. Diabetic nephropathy: endothelin antagonism for diabetic nephropathy. *Nat Rev Nephrol*. 2010;6: 447–449.
84. Inscho EW, Imig JD, Cook AK, Pollock DM. ETA and ETB receptors differentially modulate afferent and efferent arteriolar responses to endothelin. *Br J Pharmacol*. 2005;146:1019–1026.
85. Boels MG, Avramut MC, Koudijs A, et al. Atrasentan reduces albuminuria by restoring the glomerular endothelial glycocalyx barrier in diabetic nephropathy. *Diabetes*. 2016;65:2429–2439.
86. Mann JF, Green D, Jamerson K, et al. Avasentan for overt diabetic nephropathy. *J Am Soc Nephrol*. 2010;21:527–535.
87. Kohan DE, Pritchett Y, Molitch M, et al. Addition of atrasentan to renin-angiotensin system blockade reduces albuminuria in diabetic nephropathy. *J Am Soc Nephrol*. 2011;22:763–772.
88. Woroniecka KI, Park AS, Mohtat D, et al. Transcriptome analysis of human diabetic kidney disease. *Diabetes*. 2011;60:2354–2369.
89. Berthier CC, Zhang H, Schin M, et al. Enhanced expression of Janus kinase-signal transducer and activator of transcription pathway members in human diabetic nephropathy. *Diabetes*. 2009;58:469–477.
90. Marrero MB, Baner-Berceli AK, Stern DM, Eaton DC. Role of the JAK/STAT signaling pathway in diabetic nephropathy. *Am J Physiol Renal Physiol*. 2006;290:F762–F768.
91. Brosius FC, Tuttle KR, Kretzler M. JAK inhibition in the treatment of diabetic kidney disease. *Diabetologia*. 2016;59:1624–1627.
92. Reiser J, von Gersdorff G, Loos M, et al. Induction of B7-1 in podocytes is associated with nephrotic syndrome. *J Clin Invest*. 2004;113:1390–1397.
93. Benigni A, Gagliardini E, Remuzzi G. Abatacept in B7-1-positive proteinuric kidney disease. *N Engl J Med*. 2014;370:1261–1263.
94. Lin SL, Chen RH, Chen YM, et al. Pentoxifylline attenuates tubulointerstitial fibrosis by blocking Smad3/4-activated transcription and profibrogenic effects of connective tissue growth factor. *J Am Soc Nephrol*. 2005;16:2702–2713.
95. Gunduz Z, Canoz O, Per H, et al. The effects of pentoxifylline on diabetic renal changes in streptozotocin-induced diabetes mellitus. *Ren Fail*. 2004;26:597–605.
96. Davila-Esqueda ME, Martinez-Morales F. Pentoxifylline diminishes the oxidative damage to renal tissue induced by streptozotocin in the rat. *Exp Diabetes Res*. 2004;5:245–251.
97. Navarro-Gonzalez JF, Mora-Fernandez C, Muros de Fuentes M, et al. Effect of pentoxifylline on renal function and urinary albumin excretion in patients with diabetic kidney disease: the PREDIAN trial. *J Am Soc Nephrol*. 2015;26:220–229.
98. Liu D, Wang LN, Li HX, et al. Pentoxifylline plus ACEIs/ARBs for proteinuria and kidney function in chronic kidney disease: a meta-analysis. *J Int Med Res*. 2017;45:383–398.
99. Jiang X, Zhou S, Yao J, et al. Effect of pentoxifylline in proteinuric chronic kidney disease: a systematic review and meta-analysis. *J Nephrol*. 2016;29:653–662.
100. Tampe B, Tampe D, Zeisberg EM, et al. Induction of Tet3-dependent epigenetic remodeling by low-dose hydralazine attenuates progression of chronic kidney disease. *EBioMedicine*. 2015;2:19–36.
101. Smyth LJ, McKay GJ, Maxwell AP, McKnight AJ. DNA hypermethylation and DNA hypomethylation is present at different loci in chronic kidney disease. *Epigenetics*. 2014;9:366–376.
102. Ko YA, Mohtat D, Suzuki M, et al. Cytosine methylation changes in enhancer regions of core pro-fibrotic genes characterize kidney fibrosis development. *Genome Biol*. 2013;14:R108.
103. Smyth LJ, Duffy S, Maxwell AP, McKnight AJ. Genetic and epigenetic factors influencing chronic kidney disease. *Am J Physiol Renal Physiol*. 2014;307:F757–F776.
104. Tampe B, Steinle U, Tampe D, et al. Low-dose hydralazine prevents fibrosis in a murine model of acute kidney injury-to-chronic kidney disease progression. *Kidney Int*. 2017;91:157–176.